

Saad Aamir

Software Engineer · Applied ML and LLM Systems

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PROFILE

Software engineer with three years of production experience, working on two sides of applied AI: LLM systems shipped into production, and empirical ML research on whether reported results survive being measured properly.

SELECTED PROJECTS

Capitulation direction in language models 2026 · preprint, arXiv:2609.17550

[interpretability code](#) · [evaluation harness](#)

- Co-authored a preprint asking whether a model's tendency to cave under user pushback corresponds to a manipulable linear direction in the residual stream. Reported a null under a pre-registered protocol (cross-validation, shuffled-label nulls, a positive control, an AUROC gate) in which the control passed and the direction did not.
- Follow-up in preparation extends this to eight models across two families with causal ablation through TransformerLens: the direction emerges between 3B and 7B and moves behaviour only where it clears the gate comfortably.

Multi-Centre Polyp Segmentation Audit 2026 · Python, PyTorch, ONNX Runtime Web

[code](#) · [live demo](#) · [weights](#)

- Trained U-Nets with a ResNet34 encoder for colonoscopy polyp segmentation on the standard 1,450-image benchmark (1,305 training, 145 validation), then evaluated them across five clinical centres in three countries, including centres never seen during training.
- Treated the model score as a measurement: seed alone moves Dice on the hardest unseen centre by ± 0.036 , more than most improvements reported for it, so single-run comparisons are uninterpretable. Ran a pre-registered five-seed protocol over ten runs, reporting +0.055 Dice ($p = 0.024$) for augmentation.
- Found a hidden source of bias: with every tissue pixel discarded, a linear probe still identifies the source centre at 91.6% against 47.6% chance. Published the median-seed model rather than the best, and full-precision weights after int8 inflated the effect by 13.5%.

Dark Matter Co-Pilot open source · Python, MCP, Pydantic v2, SQLite

[code](#)

- Built during my freelance work to automate my own CRM and outreach: an MCP server that gives an AI assistant structured access to real operations data (leads, projects, case studies, pricing) through ten typed tools over SQLite. Pydantic v2 is the single source of truth for rows, tool schemas and responses, and I use it daily.

Fly-dysseus interactive · FlyWire connectome, NeuroMechFly, Embodied Fly Lab

[run it](#) · [code](#)

- Odysseus and the Sirens retold on a simulated fruit fly, where the Sirens are sugar and the brain is the full 138,639-neuron FlyWire connectome. Reproduced the published whole-brain model (Shiu et al., Nature 2024) to within 0.3 Hz of the authors' figures.
- Three conditions: intact; sensory neurons silenced, where the feeding circuit shifts 51 Hz and the fly ignores food; and the motor neuron cut, where the circuit stays fully active and the body walks past anyway. The same causal ablation used in LLM interpretability, on a network that evolved rather than one that was trained.

EXPERIENCE

Software Engineer (Full Stack), ZSystems (MobileLIVE, now ML Arteka), Canadian technology consultancy

Jul 2023 – Jun 2026 · Remote

- Led the backend of the company's own hiring portal: designed the API and data model, built the workflow end to end, and added three LLM-backed features on the OpenAI API. Benchmarked GPT-4 against GPT-4o and chose GPT-4o for latency.
- Requisition summarisation: the model reads a free-text job description and returns title, skills and requirements through function calling against a JSON schema I defined, so a requisition reads as a bulleted summary instead of a page of prose.
- Candidate search: the criteria the model extracts drive a query to the CATS One applicant tracking API, returning the top ten matching candidates inside the portal and replacing a manual search.
- Resume parsing and normalisation: the model maps an uploaded PDF or DOCX CV into a validated JSON schema, which is persisted and rendered in the company's standard format so any reformatted CV can be re-downloaded.
- Iterated prompts and schemas until model output was reliable enough to store, validated every response before it reached the database, and cut query latency by roughly 40%.
- Led the modernisation of two internal .NET portals for a lighting manufacturer over about a year: upgraded .NET 6 to .NET 8, introduced the repository pattern, inversion of control and dependency injection, and migrated raw SQL to LINQ.

Web Development Intern, ZSystems, Lahore Jul – Sep 2020

- Built frontends and Node.js/Express services alongside senior engineers; the company brought me back full time after I graduated in 2023.

SKILLS

Machine learning PyTorch, segmentation models, experiment design, pre-registration, significance testing, ONNX, quantisation

LLM systems OpenAI and Claude APIs, function calling, structured outputs, MCP servers, RAG, LLM-as-judge validation

Python / data Python, pandas, NumPy, SciPy, scikit-learn, SQL

Software Node.js, Express, FastAPI, React, Next.js, TypeScript, C#/.NET

Tooling Git, GitHub Actions, Docker, AWS, Vercel

EDUCATION

M.Sc. Computer Science, Technische Universität Dresden · from October 2026

B.Sc. Computer Science, National University of Sciences and Technology (NUST), Islamabad · 2019 – 2023